PULS + Deutsche Bank Grow – the finance app that grows with you

Concept for the Deutsche Bank Wealth Management challenge (Futury Build Days #4, Frankfurt, 11–12 Nov 2026): the PULS score engine plus the Grow app for Next Gen clients aged 18+, whose interface grows with life stage and competence. The core mechanic is Apply to Unlock. Includes research, methodology, legal frame, 32h plan, demo and /goal prompt.

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Summary

Two building blocks. The PULS engine is our algorithm for an explainable financial-health score from 0 to 100. Deutsche Bank Grow is the app around it for Next Gen clients aged 18+: career starters, high earners, founders and heirs who are not rich yet but on their way. Its interface grows from a Duolingo-style level path at 19 to a wealth cockpit at 32. The core mechanic is Apply to Unlock: lessons give XP, but score points and new features only come when the learning shows up in real account or portfolio data. That matches our calibration on 6,389 CFPB cases: behaviour and resilience go with wellbeing, knowledge alone does not.

53/100	52%	69 vs 76	81%

Financial wellbeing score Germany
Roland Berger

worry about money every week
Roland Berger

Financial literacy age 18–29 vs Germany overall
OECD/INFE 2023

of Next Gen HNWIs plan to switch after inheriting
Capgemini WWR 2025

Our angle
Deutsche Bank already has a static DIN 77230 score in its FinanzCheck. Grow turns it into a relationship from the first salary to the inheritance: autopilot rules, a monthly score statement, couple mode and a visible path into the wealth segment.

What does financially healthy mean?

Our answer: financially healthy means you can cover today without stress, absorb a shock, make confident decisions that move you toward your own goals, and have secured your future and the people who depend on you. It is about what you do and can do, not how much you have. A 19-year-old with €2,000 and solid habits can score higher than a careless heir.

Layer	Question	PULS dimension
Today	Do I cover my obligations and live within my means?	Financial fitness, Behaviour
Shock	Could I absorb a job loss, repair or market crash?	Financial fitness (emergency reserves), Future readiness (protection)
Steer	Do I decide confidently and stick to my plan?	Behaviour, Financial competence (literacy, confidence)
Future	Am I on track for retirement, my goals and my family?	Future readiness, Life-event readiness
Freedom	Do I have choices: independence, founding, purpose?	Financial independence, 360° profile

This follows the CFPB definition: meeting current obligations, feeling secure about

the future, and being free to make choices that let you enjoy life. Wealth is a result of financial health, not its definition.

Challenge & judging criteria

Criterion (20% each)	How PULS + Grow scores
Innovation	Apply to Unlock with proof from account data, UI that grows with life stage and competence, PULS Lab branch scanner
Relevance for Next Gen	18+ moments of truth: first salary, investing, marriage, founding, inheritance; milestones tab, couple mode
Simplicity & UX	No manual budgeting, autopilot confirmed once, one score, one next step, explanation in one sentence
Scoring methodology	Validated anchors, DIN thresholds, regression calibration, weights per life stage, data coverage as confidence
Business impact	Path from first salary into the wealth segment, retention through wealth transfer, funnel into advice and mandates

Challenge coverage

Challenge item	Status	PULS dimension / feature
Factor: financial literacy	Covered	Financial competence K1 (OECD/INFE), skill tree, certificates
Factor: financial behaviour & habits	Covered	Behaviour B1–B4, Apply to Unlock, autopilot
Factor: long-term planning	Covered	Future readiness Z1–Z3, wealth path, gap radar
Factor: entrepreneurial mindset	Strengthened	Skill path entrepreneurship, founding life event, new indicator entrepreneurial activity

Factor: financial independence	Added	New indicator independence ratio (see below)
Factor: community & networks	Added	Grow Circles and mentoring, in Learning & community
Factor: family governance & succession readiness	Strengthened	Z3 estate documents, family session, new family cockpit and indicator succession readiness
Factor: purpose & impact	Added	Impact lens in the 360° profile, shown, not scored
Financial foundation	Covered	Financial fitness F1–F3, gate
Savings habits	Covered	B1 saving months, autopilot sweep and pay-rise rule
Spend habits incl. vacation saving	Strengthened	B2 budget plus new goal spaces for planned spending (vacation pot)
Emergency reserves	Covered	F1 in months of spending, DIN 77230 target
Debt management	Covered	F3 consumer debt, B3 bills on time, overdraft guard
Investment discipline	Covered	B4, savings plan kept, no panic sale at -10%
Financial capability	Covered	Financial competence K1 + K2 self-efficacy (CFPB scale)
Financial literacy	Covered	K1, OECD/INFE knowledge and digital items
Risk awareness	Strengthened	Skill path risk & protection, scam checks, new risk-consistency signal
Confidence in decision making	Added	New indicator K3 decision confidence
Future readiness	Covered	Future readiness dimension
Retirement planning	Covered	Z1 pension coverage vs 80% of net income
Insurance protection	Covered	Z2 liability, disability (BU), health

Estate planning	Covered	Z3 power of attorney, will, emergency folder
Succession preparedness	Strengthened	New indicator Z4 succession readiness, family cockpit
Investing behaviour	Covered	B4, verification rules, no points for trading frequency
Diversification	Added	New indicator B5 diversification across all connected depots
Entrepreneurial activity	Added	New indicator side or business income and runway

Every factor and dimension from the challenge brief. Covered = already on the page; Strengthened = there before, now with its own indicator or feature; Added = new.

Closing the gaps: new features and signals

Feature	What it does	Signals	Score effect
B5 diversification	Measures concentration across all connected portfolios: largest single position, share of one stock or own employer, asset classes, regions	Holdings via portfolio connectors; targets: no position above 20%, at least 3 asset classes	Behaviour; gap radar step: diversification conversation
K3 decision confidence	3 quarterly questions (CFPB goal confidence, self-efficacy) plus behaviour: follows through on a nudge, keeps a plan through volatility	Survey; time from nudge to action; no reversal within 30 days	Financial competence; goal confidence was our second-strongest driver (+5.9)
Financial independence ratio	18–24: own income covers fixed costs, parental transfers fall. Later: passive income (dividends, interest, rent) divided by monthly	Salary vs fixed costs, transfers from parents, dividend and interest credits	Financial fitness (F4); milestone: first month fully self-financed

	spending		
Goal spaces for planned spending	Vacation pot, car, gadgets: save first, spend later instead of overdraft or buy-now-pay-later	Travel spending paid from the goal space vs overdraft or BNPL	Behaviour B2: planned spending share
Risk-consistency signal	Compares stated risk tolerance with real behaviour, plus scam-awareness checks	Panic sales, drawdown reactions, phishing quiz, fraud claims	Coaching and lessons; feeds B4
Entrepreneurial activity	Detects side income and founding, coaches runway and private/business separation	Freelance invoice credits, business account, register fees, runway in months	Life-event readiness plus 360° profile; skill path entrepreneurship
Grow Circles and mentoring	Small invite-only groups (career starters, founders, heirs) with shared goal challenges, no ranking; mentoring with DB advisors and NextGen alumni	Events attended, mentoring sessions held, circle goal completed	Learning & community (E1), capped; recognition only in the Wealth Track
Family cockpit (Z4 succession readiness)	Family charter template, family meeting with an advisor, heirs map, who-knows-what checklist, NextGen programme	Documents in the vault, family session held, beneficiaries named, heirs onboarded to Grow	Future readiness (Z4), weighted in the Wealth Track
Impact lens	Values profile and sustainability preferences, share of the portfolio in sustainable funds (SFDR Art. 8/9), recurring donations, impact goals	MiFID II sustainability preference answers, fund classification, donation transfers	Not scored: values are not a measure of health. Shown in the 360° profile and the advisor briefing

Spec v1.1: K3, B5, F4 and Z4 enter the score and weights within each dimension are renormalised. The hackathon simulator runs the 14 core indicators, so the demo numbers stay as shown. Purpose & impact and community are kept out of the 0–100 score on purpose, so nobody is penalised for their values or network.

The concept: PULS + Grow

Building block	What it is	For whom
PULS engine	Deterministic score algorithm: 6 dimensions, weights per life stage, Apply to Unlock verification rules, explanation per indicator	Runs behind every surface
Deutsche Bank Grow (standalone app)	The app that grows with you: skill tree, missions, XP, score, milestones, autopilot	Next Gen 18–35+
Grow add-on in the Deutsche Bank app	Score, statement and autopilot inside the existing DB app; later the family add-on	Existing DB clients
Advisor cockpit	Briefing from score, gaps, life events and autopilot rules; suitability assessment stays with the advisor	Next Gen advisors, Wealth Planning

PULS stands for Planen, Umsetzen, Lernen, Sichern (plan, act, learn, secure). Claim: your finances grow with you.

Apply to Unlock

1. **Lesson**
   3–5 minutes with a quiz. Gives XP and a little competence, barely any score.
2. **Behaviour signal**
   With consent, the app reads account, portfolio or transaction data.
3. **Verification rule**
   A time window and a threshold decide whether the behaviour is real and sustained.

A time window and a threshold decide whether the behaviour is real and sustained.

4 **Score points**

Only now does the engine count the points in full.

5 **Unlock**

A new feature, advisor access or the way into the next tier opens.

Tier	Lesson	Behaviour signal
Start	Budget pro	Balances, categories, overdraft days
Start	Emergency fund: 3 months	Reserve space
Start	Keep the savings plan	Savings plan at DB or an external broker
Pro	Investing 101	First savings plan of at least €150
Pro	Disability cover (BU)	Insurer direct debit or policy upload
Pro	Home-ownership readiness	Equity space
Wealth Track	Monthly sweep	Buffer after sweep
Wealth Track	Inheritance & family readiness	Family session, cover in place

Points from the engine with the weights of the respective life stage. First value = lesson only, second = with verified behaviour.

The +12.5 in Pro is the +12-points story from the brief, but with proof instead of a quiz.

Average, not cut-off date: balances count as the mean over the window, so parking money briefly earns nothing.

Round-trip detection: money in and out within 30 days and transfers between own accounts do not count as saving.

Decay: if the behaviour stops for 2 months, verification is revoked. Points drop, unlocks stay, the mission becomes active again.

Relative to income: thresholds are a percentage of net income, so small budgets are treated fairly.

Quiz points once per lesson, XP capped per month.

Data tier: only detected or confirmed data earns points; self-reported data only unlocks lessons (see life events).

Skill tree: level paths for every life stage

Skill path	Start (18–24)	Pro (25–34)	Wealth Track (30+, heirs, founders)
Financial security	Budget, emergency fund 3 months, overdraft trap	Buffer 6 months, fixed-cost ratio	Liquidity planning, cash vs invested
Investing	ETF basics, compound interest, volatility	Diversification, rebalancing, mandate vs DIY	Asset allocation, concentration risk, private markets
Wealth building	Savings rate, first €10k	Net worth, salary-raise rule	Wealth path, €250k / €500k / €2m stages
Risk & protection	Liability insurance, scams and fraud	Disability cover (BU), life insurance	Key-person and estate protection
Taxes	Tax return, saver's allowance	Capital gains, couple taxation	Inheritance and gift tax, holding structures
Retirement	State pension basics	Company pension, pension gap	80% target income, drawdown plan
Responsibility & trust	Bills on time, contracts, no buy-now-pay-later traps	Shared finances, commitments to partner	Family governance, philanthropy, stewardship
Entrepreneurship	Side hustle, freelance taxes	Founding: runway, separating private and business	Exit, succession, wealth after liquidity event

Duolingo-style: each path has nodes per life stage. A node = lesson (XP) plus a behaviour checkpoint (Apply to Unlock). The path level only rises when the checkpoint is verified.

Skill path	Behaviour checkpoint (proof)	Unlock
Financial security	Emergency fund held for 60 days	Investing module
Investing	Savings plan kept 6 months, no panic sale	Mandate info, advisor slot

Wealth building	Savings rate up 2 points for 3 months	Wealth-path forecast
Risk & protection	Policy detected via direct debit or upload	Life event: family, mortgage conversation
Taxes	Tax refund received or exemption order filed	Tax webinar certificate
Retirement	Pension statement uploaded, contribution started	Retirement cockpit
Responsibility & trust	12 months without dunning or overdraft	Couple mode, better account conditions
Entrepreneurship	Business account opened, 12-month runway saved	Founder session, business banking contact

XP and levels motivate at every age; the score measures. Lessons appear in the path where a gap is open, so adults see nudges, not homework.

Life stages 18+

Tier	Age	UI style
Grow Start	18–24	Level path with XP, badges, st
Grow Pro	25–34	Clean dashboard with mission
Grow Wealth Track	30+ / heirs, founders, high earners	Pro wealth cockpit with wealth

Weights in %	Fitness	Behaviour	Future	Life events	Competence	Learning
Start	25	30	10	10	15	10
Pro	25	30	20	10	10	5
Wealth Track	30	20	30	15	5	0

Early on behaviour and competence count most; later fitness, future and life events.

How the UI grows

Double gate	**Age and competence** A tier opens when the minimum age is reached and competence is proven (e.g. 3 verified missions or level 3). Anyone not there yet gets the next tier in guided mode.
Progressive disclosure	**Features arrive when needed** Portfolio view only after the investing lesson plus proof, succession only after a life event.
Smooth transitions	**No hard break** Each tier switch comes with a preview week. For 30 days you can go back to the old interface; navigation stays in place.
Carry-over	**History stays** XP, badges, streaks, goals, standing orders, savings plans and milestones move into every new tier. No new account, no new app.

Milestones tab & certificates

Milestone	Proof	What follows
First salary	2nd salary credit from the same employer	Pay-rise rule suggested
Emergency fund full	60-day average above target	Investing path opens
First €10k invested	Portfolio value across connected depots	Diversification lesson
Debt-free	Consumer credit and overdraft at zero for 3 months	Better conditions
First €100k net worth	Connected accounts plus manual assets	Wealth path forecast
Marriage or partnership	Confirmed life event, couple mode	Shared goals
Company founded	Business account plus register fees	Founder session

Inheritance received and planned	Large inflow parked, family session held	Wealth Planning handover

Milestones form a personal timeline and are fresh-start moments for the next nudge.

Certificates: each skill-path level and each DB webinar, seminar or NextGen event earns a certificate that can be shared, e.g. on LinkedIn.

Certificates feed the learning dimension, in full only once the linked behaviour is proven; in the Wealth Track they are recognition plus advisor access, not points.

Grow for adults (Wealth Management)

Next Gen here means: not rich yet, but on the way. Heirs, high earners and founders. For them Grow mainly measures one thing: how close am I to wealth, and what is the next step? Comparison with others is optional and off by default.

For adults fitness and future weigh more. The skill tree continues at every age, but lessons only move the score marginally; the real points come from applied behaviour.

1. Financial autopilot

Rule	Trigger	Action
Pay rise	Salary credit higher for 2 months	50% of the raise automatically into the existing savings plan (Save More Tomorrow principle)
Monthly sweep	Balance above the buffer at month end	First into the emergency fund up to 6 months of spending, then into the savings plan or mandate
Bonus / one-off payment	One-off credit from the employer	One third emergency fund, one third savings plan, one third free
Overdraft guard	Account below zero	All sweeps pause, the overdraft is repaid first
Large inflow (e.g.	Credit above a threshold	No auto-invest: parked for 90

inheritance)		days, suggestion for a wealth conversation

The app suggests rules, the client confirms once, then they run. Pausing is always one tap.

The base exists already: DB offers UltimoSparen in AktivKonto and BestKonto, moving the month-end balance to a savings account. The autopilot extends this with cascades and triggers.

Money is only invested into a savings plan or mandate the client has already chosen and that passed the suitability assessment. The app never picks products.

2. Score history as a monthly statement

October 2026	Points	Why
Start	58.0	Level 3 Stable
Emergency fund full	+3.5	3 months of spending held for 60 days
Savings plan kept	+1.2	6th execution without pausing
Overdraft used	−0.8	4 days below zero
Portfolio connected	±0	No points, but the band narrows from ±5 to ±2
End	61.9	Level 4 Confident reached

Every line comes from the engine's explain(). The language model may only put the statement into words, never invent a number.

3. Couple mode

Link	**Two profiles, one household** Both consent separately. A household score is built from shared accounts, plus joint goals such as a flat or parental leave.
Private stays private	**Sharing account by account** Each person chooses what to share: balances only, transactions too, or nothing. The individual score stays visible and personal.
Fair	**Contributions relative to income**

Pair	Contributions relative to income Joint goals show who contributes how much relative to income, without a ranking.
Unlink	Separate without losing data Each person keeps their history; shared data is simply no longer shown.

Gap radar: one gap, one step

Gap	Target	Next step
Emergency fund 2 instead of 6 months	DIN 77230: 6× net income	Switch on the monthly sweep
Pension gap 35%	80% of net income in retirement	Activate the pay-rise rule
No disability cover (BU)	80% of net income covered	BU check with an advisor
No will, no power of attorney	Documents in the vault	Start the template, 15 minutes
Concentration in one stock or company	Diversification	Request a diversification conversation

Measured against targets, not against other people. Exactly one next step per gap, sorted by points per effort.

Life events and Apply to Unlock for adults

Trigger	What the app does	What unlocks
Salary jump	Suggests the 50% rule	Held for 6 months: mandate info and advisor access
Founding	Checks runway, protection, private/business separation	Founder session, business banking contact
Home-ownership wish	Equity space with a 30% target (DIN 77230)	After 6 months of growth: mortgage conversation
Marriage, child	Couple mode, risk protection	Family cockpit

Inheritance	Parks the inflow, no marketing	Wealth Planning and family session

What unlocks are real services: advice, sessions or better conditions. Conditions never depend on credit or creditworthiness, only on proven saving and provision behaviour.

Advisor bridge and wealth path



Stage	Threshold	What the app does
Personal Banking + Grow	from €500 digital wealth management (PrivatMandat Xperience)	Autopilot, gap radar, score statement
Wealth management mandate	from €250,000 minimum investment (DB Private Banking)	At 80% of the threshold: suggests a wealth conversation
Portfolio advisory	from €500,000 (db PrivatMandat Aktiv, third-party source)	Advisor receives score, gaps and rules as preparation
Wealth Management	from €2m liquid assets (third-party source, not official)	Handover to Wealth Planning and the NextGen programme

Triggers for a wealth conversation: 80% of a threshold reached, a large inflow such as an inheritance or exit, or level 4 with 12 months of growth.

The advisor sees score, gaps, life events and autopilot rules only after the client releases them. The advisor does the suitability assessment.

For adults this wealth path is the main axis: how far to the next stage, and which rule shortens the way.

DB Private Banking: management from EUR 250,000
deutsche-bank.de

DB Wealth Management
deutsche-bank.de

DB wealth management thresholds per price list 1 Oct 2026 (third party)
geld-was-nun.de

DB raises Wealth Management floor to EUR 2m
private-banking-magazin.de · 18 Jul 2017

DB accounts compared (UltimoSparen)
deutsche-bank.de

DB WM NextGen programme
wealth.db.com

How does the app detect life events?

Event	Detectable signals	Data source	Confidence
First salary	New recurring credit with reference Lohn/Gehalt from the same employer	DB account, PSD2 AIS	High from 2nd credit
University / training	BAföG credit, semester fee to a university, trainee pay	DB account, PSD2 AIS	High
Move / first flat	New rent standing order, deposit (Kaution), removal firm, address change	DB account, master data	Medium to high
Founding	Business account opened, payments to notary, commercial register, IHK, trade office, tax adviser	DB account, PSD2 AIS	Medium, please confirm
Marriage / partnership	Joint account, name change, registry office fee	Master data, DB account	High on name change

Parenthood	Child benefit (Kindergeld) from the Familienkasse, parental allowance	DB account, PSD2 AIS	High
Inheritance	Large credit from a notary escrow or estate, payment of inheritance tax to the tax office	DB account, PSD2 AIS	Medium, confirmation only
Job loss	Salary stops, credit from the employment agency	DB account, PSD2 AIS	Medium; help, not sales

Detected	Inferred from data · Suggestion The app shows a suggestion with confidence. No points yet, no advisor alert.
Confirmed	One tap by the user · Counts fully The life-event module starts, the life-event dimension counts, missions are created.
Self-reported	Goals, plans, feelings · Lessons only Unlocks lessons but earns no score points until a hard signal follows.

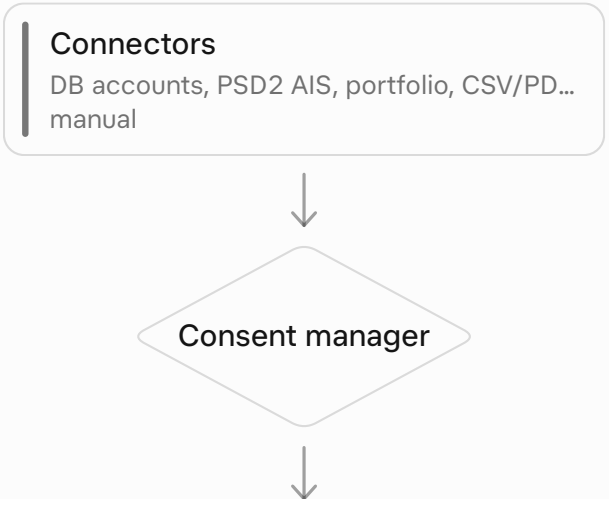
> Purpose limitation (Art. 5 GDPR): the analysis serves coaching only, never credit, pricing or sales pressure.
>
> Explicit consent for transaction analysis, revocable at any time; the app also works without it.
>
> Sensitive inferences such as pregnancy or illness are health data (Art. 9 GDPR). The CJEU (C-252/21) treats data that can reveal such information as sensitive. Therefore opt-in only, never automatic.
>
> Tact: inheritance and job loss are never opened with congratulations or a product offer.

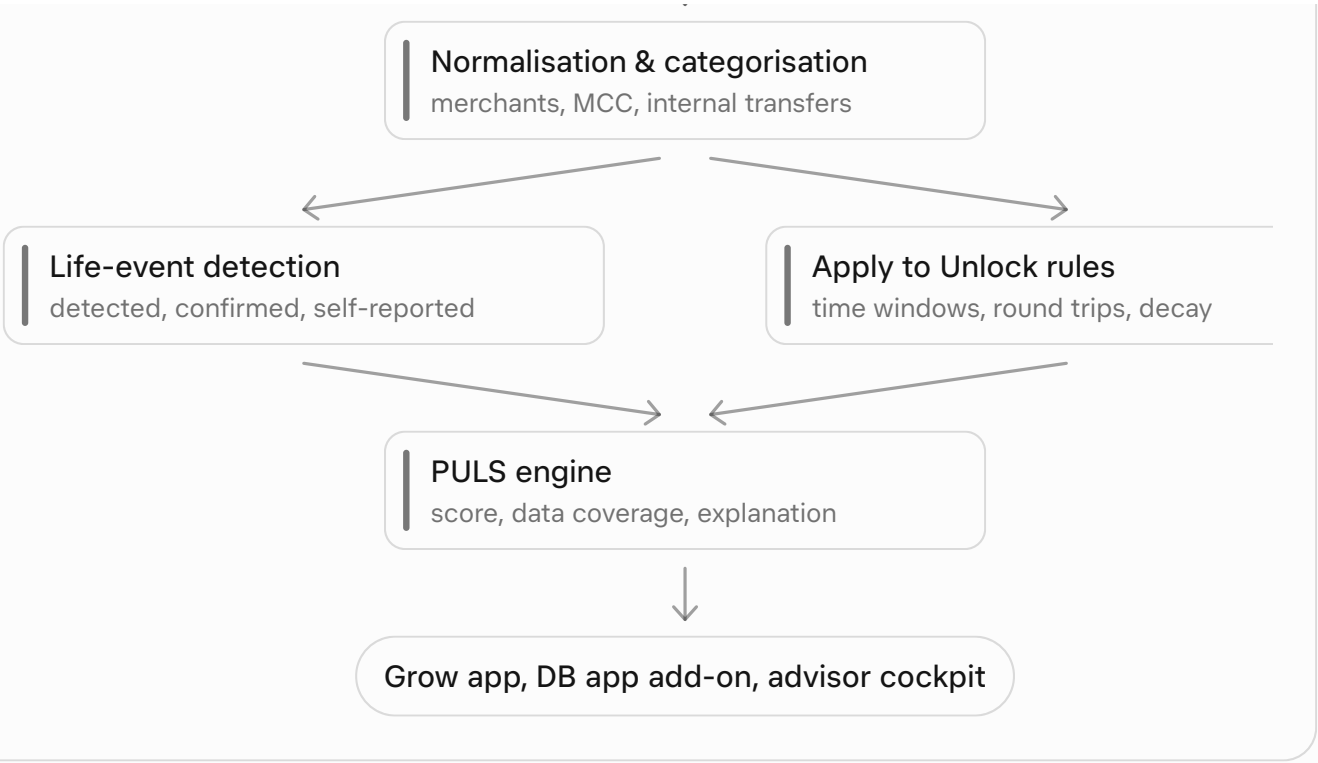
Open orchestration / multibanking

Anyone using Amex, Scalable or Trade Republic would get a distorted score from DB data alone. So Grow is open to all data sources and shows data coverage next to the

score.

Source	Access path	Status
Current accounts and cards at other banks	PSD2 account information service (AIS) via BaFin-licensed aggregators such as finAPI or Tink	Regulated, available
DB MultiBanking	The DB app already integrates accounts, cards and PayPal from other institutions	Exists
American Express (personal, Germany)	Amex offers its PSD2 API in Europe only for the UK, France, Sweden and Finland; for Germany CSV/PDF import	Gap
Portfolios (Scalable, Trade Republic)	Not covered by PSD2; finAPI connects depots via FinTS, scraping or dedicated interfaces	Available, not guaranteed by regulation
Insurance, pensions, company pension	FiDA would have covered this, but the Commission intends to withdraw it (work programme of 4 Nov 2026); hence PDF upload, e.g. the pension statement	Manual
Real estate, crypto, other	Manual entry or CSV	Manual





Data coverage	**Share of money flows PULS can see** Connected balances and flows divided by estimated total flows. A standing order to an unconnected broker is a detected gap.
Confidence	**Score with a band** From 90% high (±2 points), 70–90% medium (±5), below that low (±8, score shown as provisional).
Unlock mission	**Connect accounts** More coverage earns no points but a more precise score, and unlocks total wealth and scenarios.
DB's role	**Orchestrator, not silo** Roland Berger asks: orchestrator or solution provider? and sees open finance as the key. Grow makes DB the central finance cockpit, including money held at fintechs.

Grow simulator

Lena, 19 · Grow Start | Lena, 25 · Grow Pro | Lena, 32 · Grow Wealth Track

35.9 ± 5	**Level 2**	**70%**	**0 XP**
PULS score	Building	Data coverage	Grow level 1
Start 35.9		Confidence medium	carries over to every tier

The same person at three life stages, computed live with the engine weights of each tier; XP and unlocks carry over.

Preview Grow Start

Level 1

0 of 500 XP to the next level · path Financial security: node 2 of 5

Budget pro

Verification rule: 3 months spending below income, 0 days in overdraft · Unlock: Analytics dashboard

Lesson completed (+1.9) Behaviour proven

Emergency fund: 3 months

Verification rule: 60-day average balance above 3 months of spending · Unlock: Investing module

Lesson completed (+1.4) Behaviour proven

Keep the savings plan

Verification rule: 6 months without pausing, no panic sale at -10% · Unlock: Mandate info and advisor slot

Lesson completed (+2.4) Behaviour proven

Mission connect accounts: Scalable portfolio and Amex statement connected

Reset everything

Nothing unlocked yet: lessons give XP, unlocks only come with proof.

Scoring methodology

Dimension	Indicators (share)	Source / anchor
Financial fitness	F1 emergency fund in months of spending (40) · F2 savings rate (30) · F3 consumer debt in months of net income (30)	DIN 77230 no. 18, 19, 27; FHN Save/Borrow; Destatis
Behaviour	B1 saving months out of 12 (30) · B2 months within budget (25) · B3 bills on time (20) · B4 investment	CFPB regression; brief: reaction to volatility

	(20) · B4 investment discipline (25)	
Future readiness	Z1 pension coverage vs 80% of net (40) · Z2 basic risks liability/BU/health (35) · Z3 power of attorney, will, emergency folder (25)	DIN 77230 no. 2, 23–25, 29
Life-event readiness	L1 checklist per confirmed event	DIN 77230 no. 41; CFPB SHOCKS
Financial competence	K1 OECD knowledge + digital (60) · K2 self-efficacy (40)	OECD/INFE Toolkit 2026; CFPB FS scale
Learning & community	E1 skill-tree levels, webinars, certificates, Grow Circles, mentoring; counts half without proof	Brief: learning score; Kaiser et al. 2022

Indicator	Anchor points (value → points)
F1 emergency fund	0 → 0 · 1 month → 35 · 3 months → 70 · 6 months → 100
F2 savings rate	0% → 0 · 5% → 40 · 10% → 70 · 20% → 100
F3 consumer debt	0 → 100 · 1 month of net → 85 · 3 → 60 · 10 → 0 (DIN ceiling)
Others	linear over the share of criteria met

$$\text{PULS} = \Sigma_d \, w_{d,\text{tier}} \cdot \Sigma_i \, v_i \cdot f_i(x_i)$$

$$\Delta \text{ points} = \frac{w_d}{100} \cdot v_i \cdot (f_i(x') - f_i(x))$$

Example (Pro): emergency fund from 1 to 3 months gives 0.25 × 0.40 × (70 − 35) = +3.5 points. Forecasts always come from the engine, never from the language model.

Level (Roland Berger)	Score	Label	FHN tier

1	0–20	Fragile	Vulnerable
2	20–40	Building	Vulnerable
3	40–60	Stable	Coping
4	60–80	Confident	Coping
5	80–100	Wealth architect	Healthy

Gate	Foundation first	Emergency fund under 1 month or no personal liability insurance: level at most 2.
Peers	Percentile in your own cohort	Optional for adults and off by default. Starts with CFPB, OECD Germany and PHF benchmarks, later DB's own anonymised distribution with at least 50 people per cell.
Felt	Felt vs measured	The CFPB 5-item scale runs quarterly and separately; the gap is itself a coaching moment.
Confidence	Data coverage	The score is shown with a band when accounts are missing.

Validation	Test	Target
Construct validity	Correlation of PULS with CFPB FWB in a pilot	r ≥ 0.5
Weights	Compare with regression, sensitivity ±20%	Ranking stable
Monotonicity	No improvement lowers the score	100%
Fairness	Same behaviour, same score across gender and origin	no significant gap
Stability	Test-retest after 2 weeks	≤ 3 points

Data sources

Source	Content & variables
CFPB FWB Scale + NFWBS PUF 2016	6,394 cases, 217 variables: FWBscor GOALCONF, SHOCKS_1–11, agecat, fi
OECD/INFE Toolkit 2026	Knowledge 0–7, behaviour 0–9, attitu being 0–100
OECD/INFE 2023 + Financial Literacy in Germany	Germany: literacy 76, age 18–29 69,
Bundesbank OECD-INFE microdata	German individual data
DIN 77230:2026-03	Targets for liquidity, reserve, retireme
PHF (Bundesbank) 2023	about 4,000 households, median net
ECB HFCS	Household balance sheets, euro area
Global Findex 2025	144,090 respondents, 140+ economi
FinHealth Score + Pulse	8 indicators, 4 pillars
Melbourne Institute / CBA	Observed scale from bank data
Destatis savings rate	10.3% in H1 2025

Instant = downloadable without application. As of 11 Nov 2026.

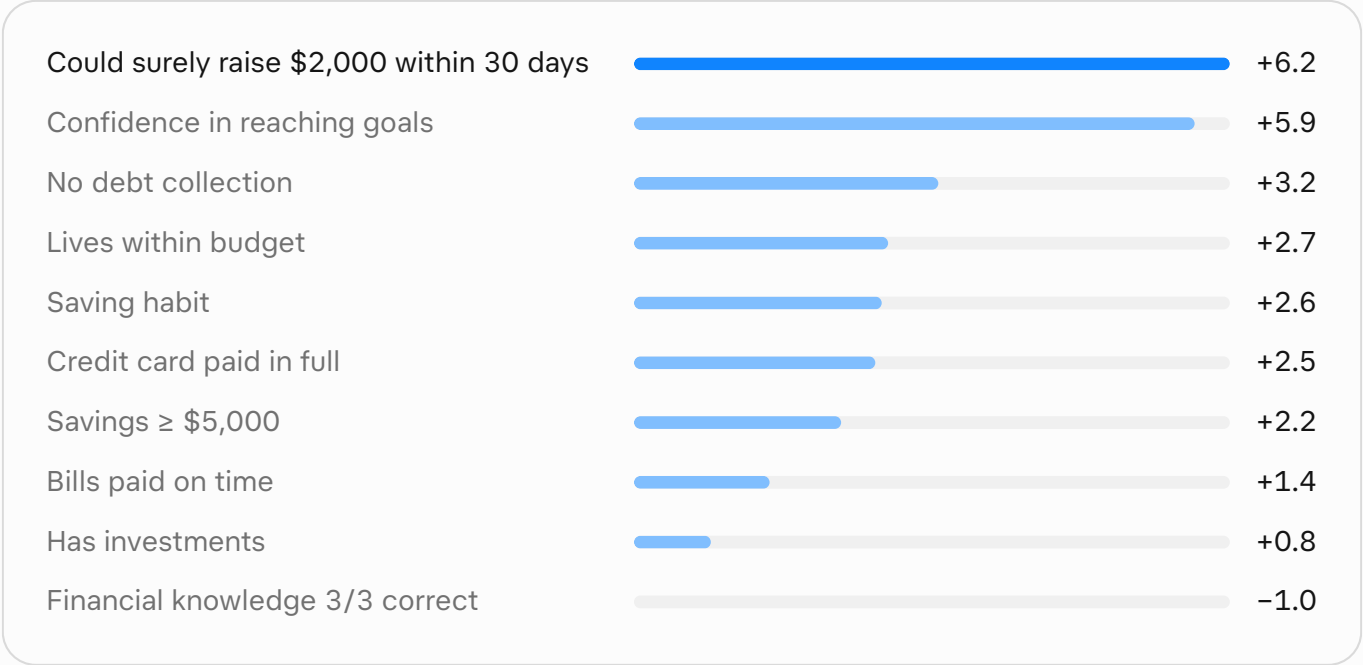
1. Top 1: CFPB scale plus NFWBS microdata, instantly analysable and already calibrated.
2. Top 2: OECD/INFE Toolkit 2026 with the 2023 Germany results.
3. Top 3: DIN 77230 targets plus PHF/HFCS tables.

Own calibration (CFPB microdata)

Age	Mean	p25	Median	p75
18–24	51.0	44	51	59
25–34	51.1	43	50	61
35–44	52.5	45	53	60
45–54	53.5	45	54	62

55–61	54.6	46	55	64
62–69	59.3	50	59	68
70–74	61.0	53	61	68
75+	59.9	50	59	69
All	54.2	46	54	63

Weighted with finalwt, n = 6,389. Cross-section, not causal.



WLS coefficients on the FWB score, controlling for income and age ($R^2 = 0.52$). Knowledge alone has no conditional effect, behaviour and resilience do. This is the empirical basis for Apply to Unlock.

Competition & gap

Provider	What it is	Gap
DB FinanzCheck	DIN 77230 analysis with a finance score and roadmap	One-off, static, no behaviour
DB Junges Konto	Free account up to age 30 for students and trainees	No learning or score layer
DB WM NextGen programme	Events, network, succession	No daily use

ANZ Financial Wellbeing Score	6 questions, score out of 100, re-check every 3 months	Quiz without account data or action
Bank of America Life Plan + Erica	Goals in 7 life areas, 21.5m plans (March 2026)	No score, little learning
CBA / Melbourne Institute	Felt plus observed scale from bank data	Research-like, little engagement
Revolut, N26, Trade Republic	Round-ups, spaces, €1 savings plans, Saveback	Saving mechanics without a holistic score
Zogo	Gamified financial literacy, white label for banks	Learning without your own finances
Finanzfluss Copilot	Aggregator, budgets, goals	No score, no advice
Sparkassen-Finanzkonzept	Advisory process, online self-check	Advisor-centric, not Next Gen

Nobody combines a lifelong score with proof, a skill tree for all ages and the path into wealth management in one interface that grows with you.

Evidence: behavioural economics

Evidence	Finding	Design
Save More Tomorrow (Thaler & Benartzi 2004)	Savings rate 3.5% → 13.6% within 40 months	Autopilot pay-rise rule
SEED commitment savings (Ashraf, Karlan & Yin 2006)	+81% balances after 12 months through commitment	Locked goal spaces
Defaults (Madrian & Shea 2001)	Defaults beat information	Autopilot rules as defaults, confirmed once
Reminders (Karlan et al. 2016)	Reminders work best with a concrete goal	Nudges always with goal and point value
Fresh start (Dai, Milkman & Riis 2014)	New beginnings raise goal behaviour	Missions at birthdays, job starts, milestones, tier switches
Kaiser, Lusardi et al	Financial education improves	Short lessons linked to an

Kaiser, Lusardi et al. 2022 (76 RCTs)	Financial education improves knowledge and behaviour	Short lessons linked to an action
Kaiser & Menkhoff 2020 (school meta-analysis)	Large effect on knowledge, small on behaviour	Learning alone is not enough: Apply to Unlock
Urban et al. 2020	Mandatory high-school financial education: better credit outcomes at 18–21	Core lessons in Grow Start
Prize-linked savings (Kearney et al.)	Lottery incentives raise saving	Monthly draw among savers, never trading prizes
Duolingo streaks	A 7-day streak makes learners 2.4× more likely to return	Weekly streaks with freeze
ESMA on gamification	Gamification that maximises trading harms investors	No points for trading frequency

Regulation & ethics

Topic	Rule	Consequence
MiFID II suitability	Investment recommendations need a suitability assessment	Grow coaches, it does not advise; products only via an advisor or the mandate journey
GDPR Art. 22 + CJEU SCHUFA (C-634/21)	A score can be an automated decision	Purpose firewall: never used for credit or pricing
GDPR Art. 9, CJEU C-252/21	Sensitive inferences are specially protected	Opt-in for sensitive categories
EU AI Act Annex III, Art. 50, Art. 4	Credit and insurance scoring are high-risk; transparency for AI	Coaching without those purposes; the coach is labelled as AI
PSD2 / PSR, ESMA retail investor journey	Account data only with consent; engagement must not override informed decisions	Consent manager; rewards for saving, learning, protection

Guardrail

The wellbeing score is not a credit score: separate data storage, no export to credit or risk systems, and the language model only verbalises what the engine computed.

PULS Lab in the branch

Scan capsule: 10 minutes at a touch screen, then a room projection of your own financial vital signs.

Live stress test: a job loss or a -30% market crash is simulated and the score visibly reacts.

Couple appointment: partners scan together and start couple mode on the spot.

Deep Scan for €500: lab-values report per DIN 77230, tax and inheritance check, family session, 12 months of coaching, re-scan after 6 months. Creditable against a mandate.

Business impact

Lever	Fact	Source
Wealth transfer	About €400bn inherited per year in Germany until 2027; about a third of young heirs switch institutions	KPMG 2026
Attrition	81% of Next Gen HNWIs plan to switch within 1–2 years of inheriting	Capgemini WWR 2025
Heir retention	4 out of 5 heirs would switch; only 7% seek will advice from their bank	FONDS professionell, 11 Nov 2026
Mandate funnel	DB aims to double discretionary wealth management in the private bank 2025–2028; PrivatMandat Xperience from €500	Deutsche Bank, 15 Oct 2026

Entry hook	Junges Konto is free for students and trainees up to 30	Deutsche Bank

Client for life: from the first salary at 19 via couple mode and founding to the inheritance, without ever switching apps.

Wealth path: Grow shows each client the distance to the next segment and hands them to the right advisor at the right moment.

Orchestrator role: multibanking makes Grow the central cockpit, including money held at fintechs.

Pilot KPIs: verified missions, autopilot rules active, data coverage, retention after tier switch, wealth conversations from the advisor bridge, mandates after suitability assessment.

Architecture & tools

Tool	Use
hessian.AI chat (kolibri-1, glm-flash)	Agent builder with RAG over the DIN matrix, RB paper and OECD toolkit as coach and lesson knowledge; scheduled chats for the monthly score statement
hessian.AI gateway (Qwen3.8-27B, Mistral-Medium-3.5, gemma-4-31B)	OpenAI-compatible API for the coach and lesson texts; key still missing, template fallback until then
Cursor + Codex / OpenClaw /goal	Build engine, verification rules, skill tree, autopilot, connector mocks and UI in parallel
Cybersecurity models	Red teaming: prompt injection, PII leaks, gaming the verification rules, realistic scam lessons
Python	calibrate.py for peer percentiles, generator for synthetic transaction streams

32h build plan

Time	Deliverable	Who

H0–2	Scope, repo, spec.md from this page, persona Lena at 19 / 25 / 32, /goal started	all
H2–8	Engine with tier weights, tests green, synthetic transaction generator	A, C
H2–8	UI skeleton with age switcher and 3 themes	B
H8–14	Apply to Unlock rules, skill tree with XP, missions with delta	A, B
H8–14	Life-event detector, milestones, data coverage, coach with guardrail	C
H14–18	Sleep in shifts, red teaming running	rotating
H18–24	Autopilot, monthly score statement, couple mode, advisor bridge, kiosk	B, C
H24–28	Polish, demo data, pitch along the 5 criteria	D + all
H28–32	Three rehearsals, backup video, submission	all

H = hours from start. Roles: A engine + verification rules, B frontend, C coach + data, D pitch.

Demo script (3 minutes)

1. **0:00 Problem**
   52% worry weekly, 81% of heirs want to switch. Banks measure once, Grow grows with you.
2. **0:20 Lena, 19, Grow Start**
   Level path. First salary detected, one tap confirms. Lesson emergency fund: XP up, score +1.4. After 60 days of proof: +13.4 points, investing module unlocked. Data coverage 70%, connect Scalable: 95%, the band shrinks.
3. **1:00 Time jump: Lena, 25, Grow Pro**
   The UI becomes a dashboard. XP, badges and milestones are still there. Investing 101 alone

The UI becomes a dashboard, XP, badges and milestones are still there. Investing 101 alone +4.1; only the proven savings plan brings +12.5 and the Pro cockpit. Couple mode with her partner, private accounts stay private.

4 **1:40 Lena, 32, Wealth Track**

Cockpit. Autopilot: 50% of each raise into the plan, monthly sweep into a 6-month buffer. Monthly statement: +3 because the emergency fund is full. Wealth path: €180k of €250k, the app proposes a wealth conversation and the advisor gets the briefing.

5 **2:20 Branch & business**

PULS Lab, Deep Scan for €500, funnel into PrivatMandat Xperience and Wealth Management; outlook: family add-on before 18.

6 **2:40 Method & guardrails**

Validated anchors, regression, weights per life stage, purpose limitation, no product picks by the app.

First /goal prompt

```
/goal Build a runnable prototype in the repo puls: the PULS score engine
plus the app 'Deutsche Bank Grow' for Next Gen clients aged 18+ (concept
demo, synthetic data only). The core mechanic is Apply to Unlock: lessons
in a Duolingo-style skill tree give XP and levels, but score points and
unlocks only come once the behaviour is proven in (synthetic) account,
portfolio and transaction data. Spec: docs/spec.md (weights per life
stage, curves, gates, missions, verification rules, skill paths from the
PULS page).

The goal is only done when ALL checks pass:
1. packages/engine (TypeScript, pure functions, no network):
score(indicators, tier), level(score, gate), whatIf(state, mission) ->
delta, explain(state) -> contributions, coverage(accounts, flows) -> data
coverage in % and confidence band.
2. packages/verify: verification rules as functions over transaction
streams (e.g. emergencyFund(stream, target, days=60) uses the average
balance, not a cut-off date). Tests: money in and out within 3 days -> no
points; held for 60 days -> verified; behaviour stopped for 2 months ->
verification revoked; internal transfers between own accounts do not count
as saving.
3. npm test green with: weights per tier sum to 100; persona Lena 25
baseline 40.9 +/- 0.1, lesson Investing 101 alone +4.1, with verified
savings plan +12.5; Lena 19 baseline 35.9, emergency-fund lesson +1.4,
verified +13.4; Lena 32 baseline 60.8; monotonicity property test (1,000
random cases); gate (emergency fund under 1 month -> level at most 2); sum
of explain() = score +/- 0.1.
4. packages/skills: skill tree with 8 paths (financial security,
investing, wealth building, risk & protection, taxes, retirement,
```

```
responsibility & trust, entrepreneurship) and nodes per life stage. Each
node has a lesson (XP) and a linked behaviour checkpoint. Test: XP rises
after a lesson, the path level only rises after the checkpoint is
verified.
5. packages/autopilot: rules pay-rise 50%, monthly sweep cascade (buffer -
> emergency fund -> plan), bonus split, overdraft guard, large inflow
parking (no auto-invest). Tests: rule runs only after one-time
confirmation; overdraft pauses all sweeps; inflow above threshold is
parked 90 days and creates an advisor suggestion; no product is ever
chosen by the app.
6. packages/lifeevents and milestones: detector on transactions. Tests: 2
salary credits from the same employer -> event first salary, status
detected, confidence >= 0.8; self-reported event does not change the
score; no inference of health data (pharmacy or baby purchases create no
event); milestones tab lists first salary, emergency fund full, first EUR
10k invested.
7. packages/connectors: adapter interface with mock adapters for DB
account, PSD2 AIS (finAPI/Tink format), portfolio (Scalable/Trade
Republic), CSV/PDF import and manual entry. Test: a transfer to an
unconnected broker lowers data coverage and creates the mission connect
accounts.
8. apps/web (Next.js + Tailwind): age switcher 19 / 25 / 32 changes theme,
navigation and features (Start: level path with XP; Pro: dashboard; Wealth
Track: cockpit with wealth path); routes /scan, /score, /skills, /missions
(lesson -> proof -> points -> unlock), /milestones, /autopilot, /statement
(monthly score statement), /couple, /advisor, /kiosk. npm run build
without errors.
9. services/coach: OpenAI-compatible client (env: LLM_BASE_URL,
LLM_API_KEY, LLM_MODEL; default hessian.AI gateway) only verbalises
explain() JSON. Test: every number in the answer appears in the JSON; no
product or buy/sell prompt; template fallback without key.
10. Playwright E2E: age 19 -> lesson emergency fund -> XP rises, score
almost unchanged -> simulate proof -> score rises exactly by the shown
delta -> age 25 -> XP and unlocks are kept, UI changes -> age 32 -> wealth
path shows distance to the EUR 250k threshold.
11. README: purpose limitation (no use for credit, pricing, insurance), AI
notice under Art. 50 AI Act, no investment advice, pre-18 family add-on is
out of MVP scope.

Limits: synthetic data only, no real bank or client data, no secrets in
the repo (.env.example instead of .env), deploy nothing. Small commits,
run npm test after each step, report the output of all checks at the end.
```

For Codex or OpenClaw. All exit criteria are machine-checkable.

Outlook / pipeline: pre-18 family add-on



Not part of the 18+ MVP. Condensed from our earlier concept work.

Parents create a Grow profile for their child in the DB app add-on; a Grow space in the parents' account takes pocket money. From 14 it converts into the child's own Junges Konto with standing orders, savings plans, history, XP and streaks carried over (never having to switch). At 18 Grow Start opens.

Early-start pension (Frühstart-Rente, planned): €10 a month from age 6 to 18 into a depot for the child, start 1 Jan 2027 (Bundestag 16 Oct 2026, Bundesrat on 20 Nov 2026).

Wish Money (Wunsch-Geld): parent-released free budget, learned automatically from a 4-week baseline and proposed 10% below it, confirmed with one tap. Need Pot (Brauch-Topf): school supplies and transport, restricted to allowed merchant categories, excluded from the bonus.

Parent matching instead of 50 cents per level: e.g. 10% on the child's own saving per verified level, capped, paid into the goal after a lock period. Overjustification research (Lepper 1973; Deci et al. 1999; Gneezy & Rustichini 2000) warns against small cash rewards; matching raised saving take-up (Duflo et al. 2006).

No public leaderboards, no loss messages, no night-time push; streak freeze automatic (DSA guidelines on minors, AI Act Art. 5(1)(b)).

Legal frame: parental consent (BGB § 107, § 1629), pocket-money rule (§ 110), no credit or overdraft without family court (§ 1643 with § 1854), GDPR Art. 8 consent by parents under 16. DB requires both parents' signatures for under-18 accounts; phone wallets at N26 and Revolut in Germany only from 16.

Business case: Wealth Management parents onboard their children, so DB knows the Next Gen long before the inheritance.

Kids/teen app	What it is	Gap
Greenlight (US)	Kids' card, chores, allowance, Level Up game, investing with parental approval	No score, ends as a kids' product

GoHenry (UK)	Card and app for 6–18, Money Missions	No proof of behaviour, no adult path
Revolut <18	Own app and card for ages 7–17, parents approve	New app at 18
N26 for under-18s	Card for 7–17 in a parent's space, child has no app	No learning, no child interface
Postbank Giro start direkt	Free up to 21, Young Money app	No learning or score layer
Step (US)	Teen card that builds credit history	US credit logic, not transferable

Sources

CFPB, OECD, FHN, Revolut and GoHenry sometimes block automated access; open them in a browser.

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